

République du Sénégal



Un Peuple – Un But – Une Foi

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UFR Sciences Appliquées et Technologies de l'Information et de la
Communication (SATIC)

Département Technologies de l'Information et de la Communication (TIC)

Master professionnelle

Systèmes D'Information (S.I)

TP2 : Contrôle des versions logicielles avec Git

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par :

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Présenté

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RAPPORT TP — CONTRÔLE DE VERSION AVEC GIT ET GITHUB

Objectif du TP

L'objectif de ce TP est de :

- Comprendre le fonctionnement de Git en local
- Manipuler les commits, branches et fusions
- Gérer les conflits
- Intégrer un dépôt local avec GitHub

Partie 1 : Initialisation du dépôt Git

Création d'un projet Git local :

```
git init
```

Création du fichier :

```
echo "I am on my way to passing the Cisco DEVASC exam" > DEVASC.txt
```

Ajout et commit :

```
git add DEVASC.txt
```

```
git commit -m "Committing DEVASC.txt to begin tracking changes"
```

```
devasc@labvm:~$ cd labs/
devasc@labvm:~/labs$ ls
devnet-src
devasc@labvm:~/labs$ git config --global user.name "Adam"
devasc@labvm:~/labs$ git config --global user.email "secknote@gmail.com"
devasc@labvm:~/labs$ git config --list
user.name=Adam
user.email=secknote@gmail.com
devasc@labvm:~/labs$ cd devnet-src/
devasc@labvm:~/labs/devnet-src$ mkdir git-intro
devasc@labvm:~/labs/devnet-src$ cd git-intro
devasc@labvm:~/labs/devnet-src/git-intro$ git init
Initialized empty Git repository in /home/devasc/labs/devnet-src/git-intro/.git/
devasc@labvm:~/labs/devnet-src/git-intro$ ls -la
.
..
.git
devasc@labvm:~/labs/devnet-src/git-intro$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
devasc@labvm:~/labs/devnet-src/git-intro$
```

```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/git-intro$
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro$ echo "I am on my way to passing the Cisco DEVASC exam" > DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ ls -la
total 16
drwxrwxr-x 3 devasc devasc 4096 May 8 15:40 .
drwxr-xr-x 15 devasc devasc 4096 May 8 15:37 ..
-rw-r--r-- 1 devasc devasc 48 May 8 15:40 DEVASC.txt
drwxrwxr-x 7 devasc devasc 4096 May 8 15:38 .git
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the Cisco DEVASC exam
devasc@labvm:~/labs/devnet-src/git-intro$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  DEVASC.txt

nothing added to commit but untracked files present (use "git add" to track)
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -m "Committing DEVASC.txt to begin tracking changes"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  DEVASC.txt

nothing added to commit but untracked files present (use "git add" to track)
devasc@labvm:~/labs/devnet-src/git-intro$ git add DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -m "Committing DEVASC.txt to begin tracking changes"
[master (root commit) d0f5bb1] Committing DEVASC.txt to begin tracking changes
1 file changed, 1 insertion(+)
create mode 100644 DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ git log
commit d0f5bb170ba0401039bc8dead39ad6da4976066f (HEAD -> master)
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:41:14 2026 +0000

    Committing DEVASC.txt to begin tracking changes
devasc@labvm:~/labs/devnet-src/git-intro$
```

Partie 2 : Modification du fichier

Ajout d'une deuxième ligne :

```
echo "I am beginning to understand Git!" >> DEVASC.txt
git add DEVASC.txt
git commit -m "Updated DEVASC.txt with second line"
```

```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/git-intro$
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro$ echo "I am beginning to understand Git!" >> DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the Cisco DEVASC exam
I am beginning to understand Git!
devasc@labvm:~/labs/devnet-src/git-intro$ git status
On branch master

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
  modified:   DEVASC.txt

no changes added to commit (use "git add" and/or "git commit -a")
devasc@labvm:~/labs/devnet-src/git-intro$ git add DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -m "Updated DEVASC.txt with second line"
[master 1d203d3] Updated DEVASC.txt with second line
1 file changed, 1 insertion(+)
devasc@labvm:~/labs/devnet-src/git-intro$ git log
commit 1d203d33675854acb766f9cb129ca5b5aa4e7faf (HEAD -> master)
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:42:46 2026 +0000

    Updated DEVASC.txt with second line

commit d0f5bb170ba0401039bc8dead39ad6da4976066f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:41:14 2026 +0000

    Committing DEVASC.txt to begin tracking changes
devasc@labvm:~/labs/devnet-src/git-intro$ git diff 1d203d33675854acb766f9cb129ca5b5aa4e7faf d0f5bb170ba0401039bc8dead39ad6da4976066f
diff --git a/DEVASC.txt b/DEVASC.txt
index 085273f..93c33fb 100644
--- a/DEVASC.txt
+++ b/DEVASC.txt
@@ -1,2 +1,3 @@
 I am on my way to passing the Cisco DEVASC exam
+I am beginning to understand Git!
devasc@labvm:~/labs/devnet-src/git-intro$
```

Partie 3 : Gestion des branches

Création et utilisation d'une branche :

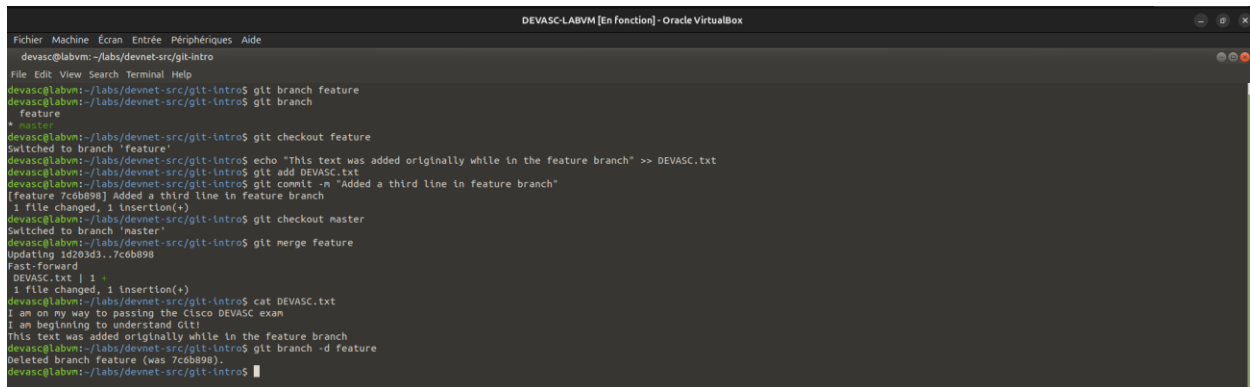
```
git branch feature  
git checkout feature
```

Modification du fichier :

```
echo "This text was added originally while in the feature branch" >>  
DEVASC.txt  
git add DEVASC.txt  
git commit -m "Added a third line in feature branch"
```

Fusion avec la branche principale :

```
git checkout master  
git merge feature  
git branch -d feature
```



```
DEVASC-LABVM [En fonction] - Oracle VirtualBox  
Fichier Machine Écran Entrée Périphériques Aide  
devasc@labvm:~/labs/devnet-src/git-intro$  
File Edit View Search Terminal Help  
devasc@labvm:~/labs/devnet-src/git-intro$ git branch feature  
devasc@labvm:~/labs/devnet-src/git-intro$ git branch  
* feature  
+ master  
devasc@labvm:~/labs/devnet-src/git-intro$ git checkout feature  
Switched to branch 'feature'  
devasc@labvm:~/labs/devnet-src/git-intro$ echo "This text was added originally while in the feature branch" >> DEVASC.txt  
devasc@labvm:~/labs/devnet-src/git-intro$ git add DEVASC.txt  
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -m "Added a third line in feature branch"  
[feature 7c6b898] Added a third line in feature branch  
1 file changed, 1 insertion(+)  
devasc@labvm:~/labs/devnet-src/git-intro$ git checkout master  
Switched to branch 'master'  
devasc@labvm:~/labs/devnet-src/git-intro$ git merge feature  
Updating 1d203d3..7c6b898  
Fast-forward  
 DEVASC.txt | 1  
+ 1 file changed, 1 insertion(+)  
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt  
I am on my way to passing the Cisco DEVASC exam  
I am beginning to understand git!  
This text was added originally while in the feature branch  
devasc@labvm:~/labs/devnet-src/git-intro$ git branch -d feature  
Deleted branch feature (was 7c6b898).  
devasc@labvm:~/labs/devnet-src/git-intro$
```

Partie 4 : Gestion des conflits

Création d'une branche de conflit :

```
git checkout -b conflict-branch
```

Modification dans la branche :

```
sed -i 's/Cisco/NetAcad/' DEVASC.txt
git commit -a -m "Changed Cisco to NetAcad"
```

Modification dans master :

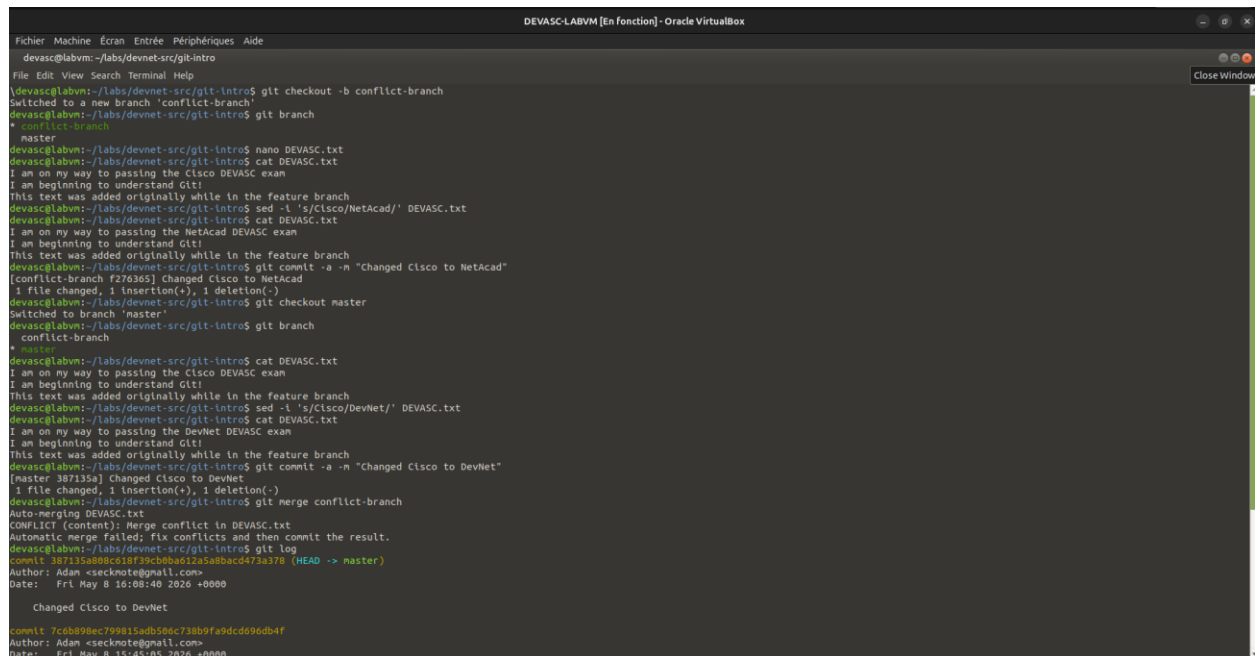
```
git checkout master
sed -i 's/Cisco/DevNet/' DEVASC.txt
git commit -a -m "Changed Cisco to DevNet"
```

Fusion provoquant un conflit :

```
git merge conflict-branch
```

Résolution manuelle du conflit :

```
vim DEVASC.txt
git commit -a -m "Manually merged from test branch"
```



```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/git-intro$
File Edit View Search Terminal Help
\devasc@labvm:~/labs/devnet-src/git-intro$ git checkout -b conflict-branch
Switched to a new branch 'conflict-branch'
devasc@labvm:~/labs/devnet-src/git-intro$ git branch
* conflict-branch
master
devasc@labvm:~/labs/devnet-src/git-intro$ nano DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the Cisco DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ sed -i 's/Cisco/NetAcad/' DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the NetAcad DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -a -m "Changed Cisco to NetAcad"
[conflict-branch f276385] Changed Cisco to NetAcad
1 file changed, 1 insertion(+), 1 deletion(-)
devasc@labvm:~/labs/devnet-src/git-intro$ git checkout master
Switched to branch 'master'
devasc@labvm:~/labs/devnet-src/git-intro$ git branch
* conflict-branch
master
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the Cisco DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ sed -i 's/Cisco/DevNet/' DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the DevNet DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -a -m "Changed Cisco to DevNet"
[master 387135a] Changed Cisco to DevNet
1 file changed, 1 insertion(+), 1 deletion(-)
devasc@labvm:~/labs/devnet-src/git-intro$ git merge conflict-branch
Auto-merging DEVASC.txt
CONFLICT (content): Merge conflict in DEVASC.txt
Automatic merge failed; fix conflicts and then commit the result.
devasc@labvm:~/labs/devnet-src/git-intro$ git log
commit 387135a988c618f39cbbba612a5a8bacd733378 (HEAD -> master)
Author: Adam <secknote@gmail.com>
Date: Fri May 8 16:08:40 2026 +0000

    Changed Cisco to DevNet

commit 7c68989bc799615adb59dc738b9fa9dc096db4f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:45:05 2026 +0000
```

```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/git-intro
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro$ git log
commit 387135a08c818f39cb0ba612a5a8bacd473a378 (HEAD -> master)
Author: Adam <secknote@gmail.com>
Date: Fri May 8 16:08:40 2026 +0000

    Changed Cisco to DevNet

commit 7c6b898ec799815adb58ec738b9fa9dcd096db4f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:45:05 2026 +0000

    Added a third line in feature branch

commit 1d283d33675854acb766f9cb129ca5b5aa4e7faf
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:42:46 2026 +0000

    Updated DEVASC.txt with second line

commit d0f5bb170ba401839bc8dead39ad6da976066f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:41:14 2026 +0000

    Committing DEVASC.txt to begin tracking changes
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
<<<<<< HEAD
I am on my way to passing the DevNet DEVASC exam
=====
I am on my way to passing the NetAcad DEVASC exam
>>>>>> conflict-branch
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ vim DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the DevNet DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$ git commit -a -m "Manually merged from test branch"
[master 09e4a00] Manually merged from test branch
```

```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/git-intro
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro$ git log
commit 09e4a003ef1b2c7f0fc737bc0241054539f00d9 (HEAD -> master)
Merge: 387135a f276365
Author: Adam <secknote@gmail.com>
Date: Fri May 8 16:14:54 2026 +0000

    Manually merged from test branch

commit 387135a08c818f39cb0ba612a5a8bacd473a378
Author: Adam <secknote@gmail.com>
Date: Fri May 8 16:08:40 2026 +0000

    Changed Cisco to DevNet

commit f27636552eb870aca9367b52f46608f27256e4ea (conflict-branch)
Author: Adam <secknote@gmail.com>
Date: Fri May 8 16:07:39 2026 +0000

    Changed Cisco to NetAcad

commit 7c6b898ec799815adb58ec738b9fa9dcd096db4f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:45:05 2026 +0000

    Added a third line in feature branch

commit 1d283d33675854acb766f9cb129ca5b5aa4e7faf
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:42:46 2026 +0000

    Updated DEVASC.txt with second line

commit d0f5bb170ba401839bc8dead39ad6da976066f
Author: Adam <secknote@gmail.com>
Date: Fri May 8 15:41:14 2026 +0000

    Committing DEVASC.txt to begin tracking changes
devasc@labvm:~/labs/devnet-src/git-intro$ cat DEVASC.txt
I am on my way to passing the DevNet DEVASC exam
=====
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro$
```

Partie 7 : Intégration avec GitHub

Création du dossier

```
mkdir devasc-study-team  
cd devasc-study-team
```

Copie du fichier

```
cp ../DEVASC.txt .
```

Initialisation du dépôt Git

```
git init
```

Configuration utilisateur :

```
git config --global user.name "Adama Seck"  
git config --global user.email "seckmote@gmail.com"
```

Lien avec GitHub

Ajout du dépôt distant :

```
git remote add origin https://github.com/Adam01-i/devasc-study-team.git  
git remote -v
```

Ajout et commit

```
git add DEVASC.txt  
git commit -m "Initial commit"
```

Push vers GitHub

```
git push origin master
```

Problème rencontré :

- Le mot de passe GitHub ne fonctionne plus

Solution :

- Utilisation d'un **Personal Access Token (PAT)** à la place du mot de passe

Résultat :

```
To https://github.com/Adam01-i/devasc-study-team.git  
* [new branch]      master -> master
```

```
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro$ mkdir devasc-study-team
devasc@labvm:~/labs/devnet-src/git-intro$ ls
devasc-study-team  DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro$ cd devasc-study-team
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ cp ../DEVASC.txt .
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ ls
DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ cat DEVASC.txt
I am on my way to passing the DevNet DEVASC exam
I am beginning to understand Git!
This text was added originally while in the feature branch
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git init
Initialized empty Git repository in /home/devasc/labs/devnet-src/git-intro/devasc-study-team/.git/
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git config --list
user.name=Adam
user.email=secknote@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git config --global user.name "Adama Seck"
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git config --global user.email "secknote@gmail.com"
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git config --list
user.name=Adama Seck
user.email=secknote@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git remote add origin https://github.com/Adam01-l/devasc-study-team.git
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git remote -v
origin https://github.com/Adam01-l/devasc-study-team.git (fetch)
origin https://github.com/Adam01-l/devasc-study-team.git (push)
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git add DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git commit -m "Initial commit"
[master (root-commit) 6690c20] Initial commit
1 file changed, 3 insertions(+)
create mode 100644 DEVASC.txt
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git log
commit 6690c200e4bd555f1b1353cb7daa09a8384528 (HEAD -> master)
Author: Adama Seck <secknote@gmail.com>
Date: Fri May 8 16:31:03 2026 +0000

    Initial commit
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git status
On branch master
nothing to commit, working tree clean
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$
```

New repository

Q Type to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
Required fields are marked with an asterisk (*).

1 General

Owner *

Adam01-l / devasc-study-team

devasc-study-team is available.

Great repository names are short and memorable. How about [crispy-bassoon](#)?

Description

Travailler ensemble pour réussir l'examen DEVASC

48 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Private

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

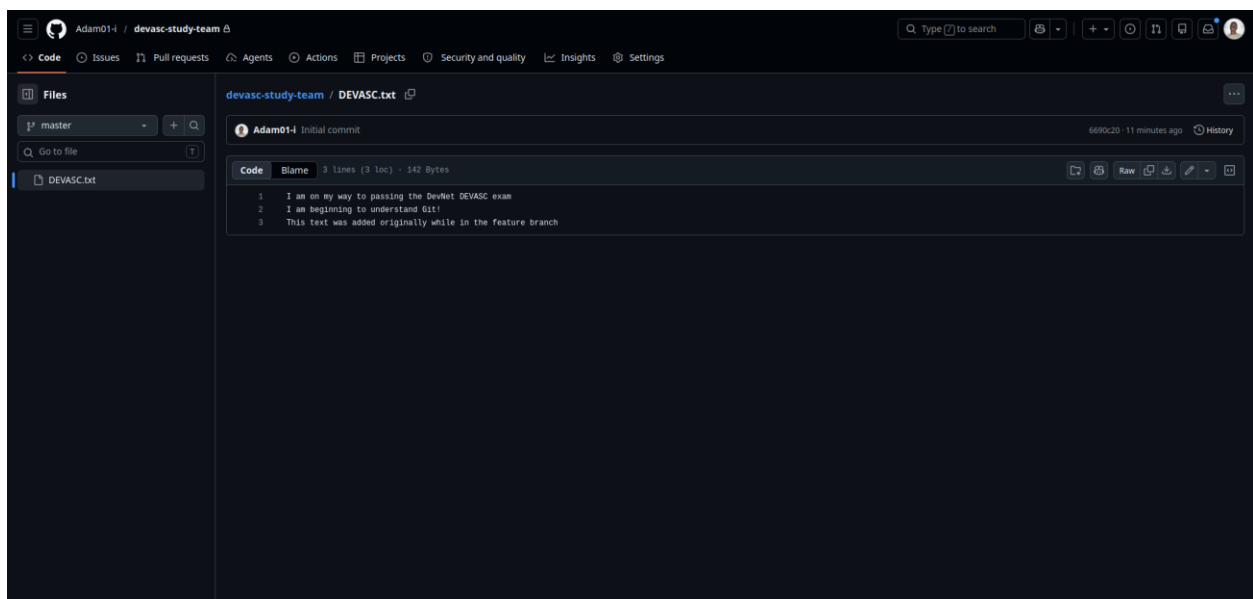
Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

```
Fichier Machine Écran Entrée Périphériques Aide
DEVASC-LABVM [En fonction] - Oracle VirtualBox
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git push origin master
Username for 'https://github.com': Adam01-t
Password for 'https://Adam01-t@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Adam01-t/devasc-study-team.git/'
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$ git push origin master
Username for 'https://github.com': Adam01-t
Password for 'https://Adam01-t@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 325 bytes | 325.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
To https://github.com/Adam01-t/devasc-study-team.git
 * [new branch]      master -> master
devasc@labvm:~/labs/devnet-src/git-intro/devasc-study-team$
```



Résultat final

- Dépôt local fonctionnel
- Gestion des branches
- Résolution de conflit
- Dépôt synchronisé avec GitHub

Conclusion

Ce TP m'a permis de :

- Comprendre les concepts fondamentaux de Git
- Manipuler les branches et résoudre des conflits
- Connecter un dépôt local à GitHub
- Utiliser un token d'authentification sécurisé